An m-covered family gives every member a certificate of size m plus one

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Editorial note. The counted statement is the author's labelled environment at source0.tex lines 581-585 and its proof environment opens at line 586; the pairing is by adjacency and the retained proof range is the whole environment.

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Editorial scope. One original result of the pinned author manuscript, counted once: the certificate bound for an m-covered family (author label lem:structure-sym).

Allowed standard prerequisites (editorial).

- m-covered family. The family is contained in the union of m cosets, each prescribing one image of a permutation.. It is the hypothesis.
- Certificate of a member. A set of images prescribed for that member; the size bound is the conclusion.. It is what the statement bounds.
- Degree reduction. Reducing a function by fixing the coordinates of a coset gives a function of lower degree on the remaining coordinates.. It is the tool that lets the degree-one classification be applied.

Lemma 0.1. *If $f\colon S_n \to \{0,1\}$ has degree at most 1 then $C(f) \leq 1$.*

Lemma 0.2. *Suppose that $f\colon S_n \to \{0,1\}$ is the characteristic function of a family which is a subset of the coset $[i \mapsto j]$. Then $\deg f|_{[i\mapsto j]} \leq \max(\deg f - 1, 0)$.*

Lemma 0.3. *If $f\colon \mathcal{M}_{2n} \to \{0,1\}$ has degree at most 1 then $C(f) \leq 2$.*

Lemma 0.4. *Let $\mathcal{F} \subseteq S_n$ be an m-covered family, say $\mathcal{F} \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$. Suppose that its characteristic function f has degree at most 2.*

Every α has a certificate of the form $\{(i_1, \alpha(i_1)), \ldots, (i_m, \alpha(i_m)), (i,j)\}$, and in particular, $C(f) \leq m+1$.

Proof. We will prove by induction on m that if $\mathcal{F} \subseteq S_n$ is m-covered, say $\mathcal{F} \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$, and its characteristic function f has degree at most 2, then for every distinct $k_1, \ldots, k_m \in [n]$, the restriction of f to $[i_1 \mapsto k_1] \cap \cdots \cap [i_m \mapsto k_m]$ has degree at most 1. By Theorem 0.1, this restriction has certificate complexity at most 1. For any $\alpha \in S_n$, choosing $k_1 = \alpha(i_1), \ldots, k_m = \alpha(i_m)$, we deduce that α has a certificate of the required form.

We start with the base case $m = 1$. Let $\mathcal{F} \subseteq S_n$ be such that $\mathcal{F} \subseteq [i_1 \mapsto j_1]$, and suppose that its characteristic function f has degree at most 2. Theorem 0.2 directly implies that the restriction of f to $[i_1 \mapsto j_1]$ has degree at most 1, as required.

Suppose now that $m \geq 2$. Let $\mathcal{F} \subseteq S_n$ be such that $\mathcal{F} \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$, and suppose that its characteristic function f has degree at most 2. Consider any distinct $k_1, \ldots, k_m \in [n]$. We need to show that the restriction of f to $[i_1 \mapsto k_1] \cap \cdots \cap [i_m \mapsto k_m]$ has degree at most 1.

Suppose first that $k_s \neq j_s$ for some $s \in [m]$, say $k_m \neq j_m$. The restriction of $\mathcal{F}$ to $[i_m \mapsto k_m]$ is contained in $[i_1 \mapsto j_1] \cup \cdots \cup [i_{m-1} \mapsto j_{m-1}]$, and its characteristic function has degree at most 2. Therefore the induction hypothesis implies that the restriction of f to $[i_1 \mapsto k_1] \cap \cdots \cap [i_m \mapsto k_m]$ has degree at most 1.

It remains to show that the restriction of f to $[i_1 \mapsto j_1] \cap \cdots \cap [i_m \mapsto j_m]$ has degree at most 1. To see this, consider the function

$$g := x_{i_1 j_1} \cdots x_{i_m j_m} f = f - \sum_{\substack{k_1,\ldots,k_m \\ (k_1,\ldots,k_m) \neq (j_1,\ldots,j_m)}} x_{i_1 k_1} \cdots x_{i_m k_m} f.$$

We showed above that the restriction of f to $[i_1 \mapsto k_1] \cap \cdots \cap [i_m \mapsto k_m]$ has degree at most 1 whenever $(k_1, \ldots, k_m) \neq (j_1, \ldots, j_m)$. Therefore every term in the sum on the right-hand side has degree at most $m+1$, since it can be represented as $x_{i_1 k_1} \cdots x_{i_m k_m}$ times the linear polynomial representing the restriction of f to $[i_1 \mapsto k_1] \cap \cdots \cap [i_m \mapsto k_m]$. Consequently, $\deg g \leq \max(\deg f, m+1) = m+1$.

By construction, g is the characteristic function of a family which is contained in the coset intersection $\mathcal{I} = [i_1 \mapsto j_1] \cap \cdots \cap [i_m \mapsto j_m]$. Applying Theorem 0.2 repeatedly, we conclude that the restriction of g to $\mathcal{I}$ has degree at most $\max(m+1-m, 0) = 1$. Since the restriction of f to $\mathcal{I}$ is the same as the restriction of g to $\mathcal{I}$, this completes the proof of the induction step. □

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